6 Oct 2026 22:31:47 - Sample_Calculations_Chapter4.sm
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Sample calculations, Chapter 4 (Experimental Design): one worked example of each calculation type

1. Displacement, mass and ballast (Section 4.7.1)

$\Delta := 264.87\ \mathrm{N}$ — *Displacement weight*

$g := 9.81\ \frac{\mathrm{m}}{\mathrm{s}^2}$ — *Gravitational acceleration*

$m := \frac{\Delta}{g} = 27\ \mathrm{kg}$ — *Displaced mass*

$m_{bare} := 11.99\ \mathrm{kg}$ — *Bare hull mass (scale)*

$m_{ballast} := m - m_{bare} = 15.01\ \mathrm{kg}$ — *Ballast needed*

2. Water density, displaced volume and Fn∇ (Equations 4.1 and 4.2)

Fresh water at 15.8 °C, ITTC (2011) Appendix A

$\rho := 998.9783\ \frac{\mathrm{kg}}{\mathrm{m}^3}$

$Vol := \frac{\Delta}{\rho \cdot g} = 0.027\ \mathrm{m}^3$ — *Displaced volume ∇ (Equation 4.2)*

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√(g ∇^(1/3)), denominator of Equation 4.1

$$V_{ref} := \sqrt{g \cdot Vol^{\left(\frac{1}{3}\right)}} = 1.7158 \ \frac{m}{s}$$

Mean measured speed, step 5 (three sets)

$$V := 2.47329 \ \frac{m}{s}$$

$$Fn := \frac{V}{V_{ref}} = 1.4415$$ *Fn ∇ (Equation 4.1)*

3. Reduction of one run to trim (Equation 4.3): 2.47 m/s, window 14.0 to 26.0 s

$R_{bow} := -37.7925$ mm *Mean bow reading*

$R_{stern} := 36.9525$ mm *Mean stern reading*

$\delta_{bow} := -R_{bow} = 0.0378$ m *Bow rise = − reading*

$\delta_{stern} := -R_{stern} = -0.037$ m *Stern rise = − reading*

$d_{pot} := 1690$ mm *Attachment spacing*

$\tau_0 := 2.2$ *Static floating trim (deg)*

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$$\tau := \tau_0 + \frac{\operatorname{atan}\left(\frac{\delta_{bow} - \delta_{stern}}{d_{pot}}\right) \cdot 180}{\pi} = 4.7324$$

Running trim (deg)

4. Uncertainty of a mean (Equations 4.5 and 4.6): 2.47 m/s

$n_{sets} := 3$ *Number of sets*

Student's t, 95 %, 2 degrees of freedom

$t_{95} := 4.303$

$F_1 := 32.307\ \text{N}$ *Set A mean drag*

$F_2 := 32.2677\ \text{N}$ *Set B mean drag*

$F_3 := 32.2184\ \text{N}$ *Set C mean drag*

$$F_{avg} := \frac{F_1 + F_2 + F_3}{n_{sets}} = 32.2644\ \text{N}$$

Mean of the three sets

Standard deviation of the set means

$$s_F := \sqrt{\frac{\left(F_1 - F_{avg}\right)^2 + \left(F_2 - F_{avg}\right)^2 + \left(F_3 - F_{avg}\right)^2}{n_{sets} - 1}} = 0.0444\ \text{N}$$

$$s_{Favg} := \frac{s_F}{\sqrt{n_{sets}}} = 0.0256\ \text{N}$$

s / √N (Equation 4.6)

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$b_F := 0\ \mathrm{N}$ *No confirmed load-cell accuracy*

Expanded uncertainty (Equation 4.5)

$u_F := t_{95} \cdot \sqrt{b_F^{\,2} + s_{Favg}^{\,2}} = 0.1103\ \mathrm{N}$

Uncertainty as % of the mean

$u_{rel} := \frac{u_F}{F_{avg}} \cdot 100 = 0.3418$

$\tau_1 := 4.73373$ *Set A mean trim (deg)*

$\tau_2 := 4.66913$ *Set B mean trim (deg)*

$\tau_3 := 4.65864$ *Set C mean trim (deg)*

Mean trim (deg)

$\tau_{avg} := \frac{\tau_1 + \tau_2 + \tau_3}{n_{sets}} = 4.6872$

Standard deviation (deg)

$s_\tau := \sqrt{\frac{\left(\tau_1 - \tau_{avg}\right)^2 + \left(\tau_2 - \tau_{avg}\right)^2 + \left(\tau_3 - \tau_{avg}\right)^2}{n_{sets} - 1}} = 0.0407$

s / √N (deg)

$s_{\tau avg} := \frac{s_\tau}{\sqrt{n_{sets}}} = 0.0235$

$r_{pot} := 0.2\ \mathrm{mm}$ *Potentiometer resolution (P25)*

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Half resolution per potentiometer through Equation 4.3 (deg)

$$b_\tau := \frac{\frac{\sqrt{2} \cdot \frac{r_{pot}}{2}}{d_{pot}} \cdot 180}{\pi} = 0.0048$$

Expanded uncertainty (deg)

$$u_\tau := t_{95} \cdot \sqrt{b_\tau^{\,2} + s_{\tau avg}^{\,2}} = 0.1031$$

5. Signal-to-noise ratio and standard error (Section 4.6): drag, 2.47 m/s run

Mean drag in the 14.0 to 26.0 s window

$$F_w := 32.2853 \text{ N}$$

Standard deviation of the 300 Hz signal

$$s_w := 0.629111 \text{ N}$$

Signal-to-noise ratio

$$SNR := \frac{F_w}{s_w} = 51.3189$$

Samples in the window (12 s × 300 Hz)

$$n_w := 3600$$

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$$SE := \frac{s_w}{\sqrt{n_w}} = 0.0105\ \text{N}$$

Standard error of the run mean

SE as a fraction of the set-to-set s (Section 4)

$$q := \frac{SE}{s_F} = 0.2362$$

6. Potentiometer linearity propagated to trim (Section 4.11): P25 variant

$$FSO := 750\ \text{mm}$$

Full-scale output, WPS-750

$$e_{lin} := 0.0025 \cdot FSO = 0.0019\ \text{m}$$

Linearity ±0.25 % FSO

Trim bound, both potentiometers (deg)

$$B_{lin} := \frac{\frac{\sqrt{2} \cdot e_{lin}}{d_{pot}} \cdot 180}{\pi} = 0.0899$$

$$b_{lin} := \frac{B_{lin}}{2} = 0.0449$$

Standard uncertainty B/2 (deg)

7. Load-cell range used (§4.11)

Largest mean drag (480 mm, 3.48 m/s)

$$F_{max} := 57.5235\ \text{N}$$

$$F_{cap} := 490.5\ \text{N}$$

Load-cell capacity

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$use := \frac{F_{max}}{F_{cap}} \cdot 100 = 11.7275$ *Percentage of capacity*

8. Reynolds number and laminar length (Section 4.11, Equation 4.4): 0.41 m/s, 15.8 °C

$\nu := 1.115 \cdot 10^{-6} \frac{m^2}{s} = 1.115 \cdot 10^{-6}\ s\ Gy$ *ITTC (2011) at 15.8 °C*

$L_p := 1.6\ m$ *Planing length*

$V_{lo} := 0.41 \frac{m}{s}$ *Speed*

$Re := \frac{V_{lo} \cdot L_p}{\nu} = 5.8834 \cdot 10^{5}$ *Model Reynolds number*

Flat-plate transition (White, 1999)

$Re_{tr} := 5 \cdot 10^{5} = 5 \cdot 10^{5}$

$x_{tr} := \frac{Re_{tr} \cdot \nu}{V_{lo}} = 1.3598\ m$ *Laminar length (Equation 4.4)*

9. Trim-dependent drag term at the trim peak (Section 4.13): 480 mm

$\tau_{pk} := 9.8768$ *Trim peak (deg), 3.48 m/s*

$D_{pk} := 57.5235\ N$ *Drag at the trim peak*

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$$share := \frac{\Delta \cdot \tan\left(\frac{\tau_{pk} \cdot \pi}{180}\right)}{D_{pk}} \cdot 100 = 80.1702$$

Δ tan τ as % of the measured drag

10. Wetted area (Equations 4.7 and 4.8): LCG 560 mm, at 2.04 m/s

$b := 0.4\ \text{m}$ *Beam*

$\beta := 18$ *Deadrise angle (deg)*

$L_k := 1.585\ \text{m}$ *Measured wetted keel length*

$L_c := 1.365\ \text{m}$ *Measured wetted chine length*

Rectangle aft of the chine points, b Lc

$A_{rect} := b \cdot L_c = 0.546\ \text{m}^2$

$A_{tri} := 0.5 \cdot b \cdot \left(L_k - L_c\right) = 0.044\ \text{m}^2$ *Two triangles, ½ b (Lk − Lc)*

Plan-view wetted area (Equation 4.7)

$A := A_{rect} + A_{tri} = 0.59\ \text{m}^2$

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Wetted bottom surface (Equation 4.8)

$$S := \frac{A}{\cos\left(\frac{\beta \cdot \pi}{180}\right)} = 0.6204\ \mathrm{m}^2$$

Mean wetted length-to-beam ratio (Equation 4.9)

$$\lambda := \frac{L_k + L_c}{2 \cdot b} = 3.6875$$

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